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Thermodynamic assessments of the Co–Er and V–Er systems

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ABSTRACT

The Co–Er and V–Er binary systems have been thermodynamically assessed by using the CALPHAD (calculation of phase diagrams) approach on the basis of the experimental data including the thermodynamic properties and phase equilibria. Gibbs free energies of the solution phases (liquid, fcc, bcc and hcp) were modeled by the subregular solution model with the Redlich–Kister formula, and those of the intermetallic compounds ($Co_{17}Er_2$, Co_5Er, Co_7Er_2, Co_3Er, Co_2Er, Co_7Er_{12} and $CoEr_3$) were described by the sublattice model. A proper set of the thermodynamic parameters has been derived for describing the Gibbs free energies of each phase in the Co–Er and V–Er systems. An agreement between the calculated results and experimental data is obtained.

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1. Introduction

In recent years, the Co–V base alloys are considered to be important potential materials in applications of high-density recording media [1], because there exists the ferromagnetic phase (fcc_{ferro} or hcp_{ferro}) and paramagnetic phase (fcc_{para} or hcp_{para}) separation induced by magnetic transformation in the Co-rich portion [2–4], which is similar to that in the Co–Cr system [5]. In order to develop the Co–V base alloys for the application of new magnetic materials, the alloying elements are used to improve the properties of magnetic recording media materials [6,7]. Knowledge of phase equilibria is important for understanding of the roles of alloying elements. The thermodynamic assessments of the Co–Er and V–Er systems which forms a portion of the Co–V base thermodynamic database are necessary.

The purpose of this work is to carry out the thermodynamic assessments of the Co–Er and V–Er systems by means of the CALPHAD (calculation of phase diagrams) method [8], in which the Gibbs free energy of each phase is described by a thermodynamic model. The thermodynamic parameters of each phase in the Co–Er and V–Er systems are optimized by fitting the experimental data on thermodynamic properties and phase equilibria.

2. Thermodynamic models

The information on stable solid phases and the used models in the Co–Er and V–Er systems is listed in Table 1.

2.1. Solution phases

The Gibbs free energies of the solution phases (liquid, fcc, bcc and hcp) in an *A–B* system corresponding to the Co–Er and Er–V systems are described by the subregular solution model, as follows:

$$G_m^{\phi} = \sum_{i=A,B} {}^0G_i^{\phi} x_i + RT \sum_{i=A,B} x_i \ \ln x_i + \Delta^E G_m^{\phi} + \Delta^{mag} G_m^{\phi}, \quad (1)$$

where ${}^0G_i^{\phi}$ is the Gibbs free energy of the pure component *i* in the respective reference state with the ϕ phase, which is taken from the SGTE pure element database [9]; x_i denotes the molar fraction of the component *i*; *R* is the gas constant; *T* is the absolute temperature; and $\Delta^E G_m^{\phi}$ is the excess Gibbs free energy, which is expressed in the Redlich–Kister polynomial [10] by:

$$\Delta^E G_m^{\phi} = x_A x_B \sum_{i=0}^{n} {}^iL_{A,B}^{\phi} (x_A - x_B)^i, \quad (2)$$

with

$${}^iL_{A,B}^{\phi} = a + bT \quad (3)$$

where ${}^iL_{A,B}^{\phi}$ is the binary interaction parameter, and the coefficients of *a* and *b* are evaluated on the basis of the available experimental data.

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Table 1
The stable solid phases and the used models in the Co–Er and V–Er systems.

System	Phase	Strukturbericht designation	Prototype	Modeling phase	Used models
Co–Er	(αCo)	A1	Cu	fcc	Subregular solution model
	(εCo)	A3	Mg	hcp	Subregular solution model
	$Co_{17}Er_2$	–	$Ni_{17}Th_2$	$(Co)_{17}(Er)_2$	Sublattice model
	Co_5Er	$D2_d$	$CaCu_5$	$(Co)_5(Er)$	Sublattice model
	Co_7Er_2	–	Co_7Er_2	$(Co)_7(Er)_2$	Sublattice model
	Co_3Er	–	Be_3Nb	$(Co)_3(Er)$	Sublattice model
	Co_2Er	C15	Cu_2Mg	$(Co)_2(Er)$	Sublattice model
	Co_7Er_{12}	–	Co_7Ho_{12}	$(Co)_7(Er)_{12}$	Sublattice model
	$CoEr_3$	$D0_{11}$	Fe_3C	$(Co)(Er)_3$	Sublattice model
	(Er)	A3	Mg	hcp	Subregular solution model
V–Er	(V)	A2	W	bcc	Subregular solution model
	(Er)	A3	Mg	hcp	Subregular solution model

Because there are the magnetic transformations in the Co_5Er and Co_7Er_2 compounds, the magnetic contributions to Gibbs free energies for these compounds should be considered. $\Delta^{mag}G_m^{\phi}$ is the magnetic contribution to the Gibbs free energy, which is described by the following equation [11]:

$$\Delta^{mag}G_m^{\phi} = RT\ \ln(\beta+1)f(\tau), \tag{4}$$

where $\tau = T/T_C^{\phi}$, T_C^{ϕ} is the Curie temperature of solution for ferromagnetic ordering and β is the Bohr magneton number, the function $f(\tau)$ is formulated by the polynomial of the normalized temperature, as follows:

$$f(\tau) = 1 - \frac{\left[79\tau^{-1}/140p + 474/497((1/p)-1)(\tau^3/6+\tau^9/135+\tau^{15}/600)\right]}{D} \quad \text{for } \tau \le 1 \tag{5}$$

$$f(\tau) = -\frac{(\tau^{-5}/10)+(\tau^{-15}/315)+(\tau^{-25}/1500)}{D} \quad \text{for } \tau > 1 \tag{6}$$

where $D = (518/1125) + (11692/15975)((1/p)-1)$, p depends on the structure, 0.4 for bcc structure and 0.28 for others.

2.2. Stoichiometric intermetallic compounds

The intermetallic compounds of $Co_{17}Er_2$, Co_5Er, Co_7Er_2, Co_3Er, Co_2Er, Co_7Er_{12} and $CoEr_3$ in the Co–Er system are treated as stoichiometric phases. The Gibbs free energy for per mole of formula unit $(Co)_s(Er)_t$ can be expressed by the two-sublattice model, as follows:

$$\begin{aligned}\Delta^0 G_f^{(Co)_s(Er)_t} &= {}^0G_m^{(Co)_s(Er)_t} + \Delta^{mag}G_m^{(Co)_s(Er)_t} - s\,{}^0G_{Co}^{SER} - t\,{}^0G_{Er}^{SER} \\ &= a' + b'T + c'T\ \ln T,\end{aligned} \tag{7}$$

where $\Delta^0 G_f^{(Co)_s(Er)_t}$ represents the Gibbs free energy of formulation per mole of formula unit $(Co)_s(Er)_t$ referred to the standard element reference (SER) state of the component elements. The parameters of a', b' and c' are optimized in the present work.

$\Delta^{mag}G_m^{(Co)_s(Er)_t}$ is the magnetic contribution to ${}^0G_m^{(Co)_s(Er)_t}$, which is expressed by Eq. (4), where the values of β and T_C^{ϕ} of each intermetallic compound are taken from the literature [12]. Since there is no magnetic transformation in the V–Er alloy, thus, the magnetic contribution to the Gibbs free energy is not considered in the V–Er system.

3. Experimental information

3.1. The Co–Er system

The phase equilibria and the intermetallic compounds in the Co–Er system have been studied by a number of investigators [13–19]. However, the reports on the existence of some compounds and relevant phase relationship by different investigators are inconsistent. There are two versions of the phase diagram in the Co–Er system proposed by Buschow [16] and Wu et al. [19], respectively. The major differences between them are as follows: (1) Wu et al. [19] and Buschow [16] reported the congruent melting point of the $Co_{17}Er_2$ compound is about 1442 °C and 1355 °C, respectively; (2) Wu et al. observed that the Co_3Er_4 phase is metastable phase [19], while Buschow included the Co_3Er_4 phase in the stable phase diagram [16]; (3) Wu et al. identified that the phase near 64 at.% Er is the Co_9Er_{16} phase [19], rather than the Co_7Er_{12} phase by Buschow [16]. However, Okamoto and Massalski [20] pointed out that the shape of the liquidus related to the Co_7Er_2 phase shown by Wu et al. [19] is quite sharp, which questioned the phase stability at lower temperatures. And Okamoto [21] also pointed out that the Co_7Er_{12} compound might be the correct stoichiometry, because the Co_9Er_{16} compound was obtained from XRD measurements [19], whereas the Co_7Er_{12} compound was based on the crystal structure data [22]. Based on all available experimental information, Okamoto [21] reviewed the phase diagram of the Co–Er system, as shown in Fig. 1.

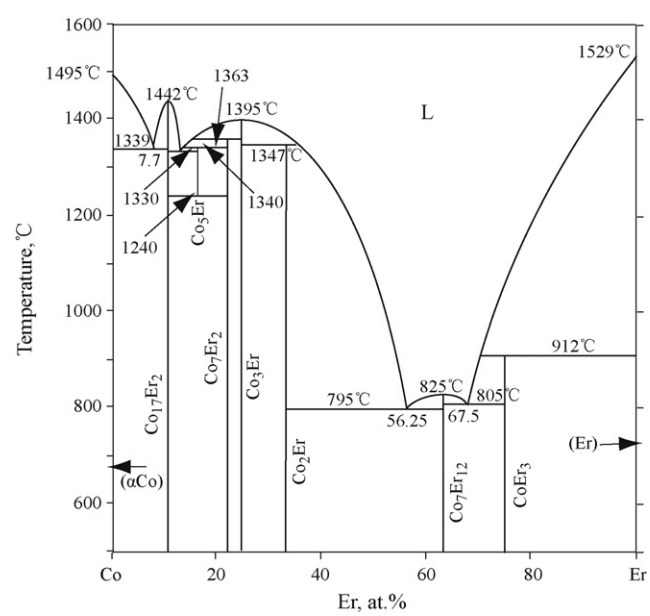

Fig. 1. The phase diagram of the Co–Er system reviewed by Okamoto [21].

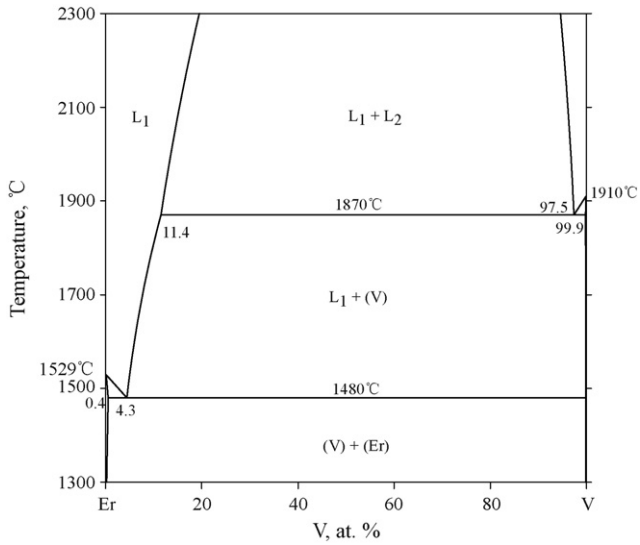

Fig. 2. The phase diagram of the V–Er system reviewed by Smith [29].

Only a few experimental data are available for the enthalpy of formation of intermetallic compounds in the Co–Er system. Schott and Sommer [23] determined the enthalpy of formation of the intermetallic phases of Co_3Er, Co_2Er and $CoEr_3$ at 825 °C by solution calorimetry.

3.2. The V–Er system

Love [24] reported the extensive miscibility gaps in both liquid and solid phases in the V–Er system with a monotectic reaction ($L_2 \leftrightarrow (V) + L_1$) near the melting temperature of V (1910 °C) and a eutectic reaction ($L \leftrightarrow (V) + (Er)$) below the melting temperature of Er (1529 °C). In later studies, many researchers confirmed that there is no intermetallic compound in this system, but gave the different temperatures and compositions of the invariant reactions [25–28]. According to all available experimental information, Smith and Lee [29] reviewed the phase diagram in the V–Er system, as showed in Fig. 2, and evaluated the interaction parameters in the V–Er system. But these parameters for expressing the Gibbs free energies of phases are not in agreement with the SGTE pure element database [9], and these parameters cannot be compatible with those in other systems. Thus, it is necessary to reassessment of the phase diagram in the V–Er system.

4. Optimization

The optimization of the thermodynamic parameters was carried out by using the PARROT module in Thermo-Calc software [30]. This software allows the introduction of a great variety of experimental data in the evaluation. The program operates by minimizing the square of the sum of errors between the calculated values and experimental data. A statistical weight assigned to each experimental data according to its compatibility with the other ones was changed by trial and error during the assessment.

All optimized parameters in the Co–Er and V–Er binary systems are listed in Tables 2 and 3, respectively.

5. Calculated results and discussion

5.1. The Co–Er system

The calculated Co–Er phase diagram compared with the experimental data is presented in Fig. 3. The calculated results are in agreement with most of the experimental data. However, the calculated melting temperature of the $Co_{17}Er_2$ compound is 1372 °C, which is slightly lower than 1442 °C proposed by Wu et al. [19], but is higher than 1355 °C by Buschow [16].

Table 2
Optimized thermodynamic parameters of the Co-Er system.

Parameters in each phase (J/mol)
Liquid phase: (Co, Er)
${}^0L^{Liq}_{Co,Er} = -103,916 + 29.404T$
${}^1L^{Liq}_{Co,Er} = +10,100 - 15.57T$
${}^2L^{Liq}_{Co,Er} = -11,500 + 5.84T$
Hcp phase: (Co, Er)
${}^0L^{Hcp}_{Co,Er} = +20,000$
$T^{Hcp}_{C,Co} = 1396$
$\beta^{Hcp}_{Co} = 1.35$
Fcc phase: (Co, Er)
${}^0L^{Fcc}_{Co,Er} = +28,000$
$T^{Fcc}_{C,Co} = 1396$
$\beta^{Fcc}_{Co} = 1.35$
$Co_{17}Er_2$ compound: $(Co)_{17}(Er)_2$
${}^0G^{Co_{17}Er_2}_f - 17{}^0G^{Hcp}_{Co} - 2{}^0G^{Hcp}_{Er} = -251,095 + 29.91T$
$T^{Co_{17}Er_2}_C = 1184$ Ref. [12]
$\beta^{Co_{17}Er_2} = 10$ Ref. [12]
Co_5Er compound: $(Co)_5(Er)$
${}^0G^{Co_5Er}_f - 5{}^0G^{Hcp}_{Co} - {}^0G^{Hcp}_{Er} = -99,101 + 7.53T$
$T^{Co_5Er}_C = 1053$ Ref. [12]
$\beta^{Co_5Er} = 1.28$ Ref. [12]
Co_7Er_2 compound: $(Co)_7(Er)_2$
${}^0G^{Co_7Er_2}_f - 7{}^0G^{Hcp}_{Co} - 2{}^0G^{Hcp}_{Er} = -193,942 + 20.854T$
$T^{Co_7Er_2}_C = 646$ Ref. [12]
$\beta^{Co_7Er_2} = 7.5$ Ref. [12]
Co_3Er Compound: $(Co)_3(Er)$
${}^0G^{Co_3Er}_f - 3{}^0G^{Hcp}_{Co} - {}^0G^{Hcp}_{Er} = -90,885 + 8.36T$
$T^{Co_3Er}_C = 395$ Ref. [12]
$\beta^{Co_3Er} = 4.2$ Ref. [12]
Co_2Er Compound: $(Co)_2(Er)$
${}^0G^{Co_2Er}_f - 2{}^0G^{Hcp}_{Co} - {}^0G^{Hcp}_{Er} = -65,972 + 1.63T$
$T^{Co_2Er}_C = 38.3$ Ref. [12]
$\beta^{Co_2Er} = 6.02$ Ref. [12]
Co_7Er_{12} compound: $(Co)_7(Er)_{12}$
${}^0G^{Co_7Er_{12}}_f - 7{}^0G^{Hcp}_{Co} - 12{}^0G^{Hcp}_{Er} = -296,500 + 4.5T$
$T^{Co_7Er_{12}}_C = 25$ Ref. [12]
$\beta^{Co_7Er_{12}} = 5.8$ Ref. [12]
$CoEr_3$ compound: $(Co)(Er)_3$
${}^0G^{CoEr_3}_f - {}^0G^{Hcp}_{Co} - 3{}^0G^{Hcp}_{Er} = -41,807 + 1.231T - 0.7T \ln T$
$T^{CoEr_3}_C = 7$ Ref. [12]
$\beta^{CoEr_3} = 6.9$ Ref. [12]

Table 3
Optimized thermodynamic parameters of the V–Er system.

Parameters in each phase (J/mol)
Liquid phase: (Er, V)
${}^0L^{Liq}_{Er,V} = +55,550$
${}^1L^{Liq}_{Er,V} = -12,750$
Bcc phase: (Er, V)
${}^0L^{Bcc}_{Er,V} = +84,000 + 21T$
Hcp phase: (Er, V)
${}^0L^{Hcp}_{Er,V} = +67,000$

Table 4
Calculated special points in the Co–Er system with the experimental data.

Reaction	Reaction type	Composition at% Er			T (°C)	Ref.
$L \leftrightarrow (\alpha Co) + Co_{17}Er_2$	Eutectic	6.5	0	10.5	1347	This work
		7.7	0	10.5	1339	[19]
$L \leftrightarrow Co_{17}Er_2$	Congruent melting		10.5		1372	This work
			10.5		~1442	[19]
$L \leftrightarrow Co_{17}Er_2 + Co_5Er$	Eutectic	16	10.5	16.7	1336	This work
		13.5	10.5	16.7	1330	[19]
$L + Co_7Er_2 \leftrightarrow Co_5Er$	Peritectic	16.7	22.2	16.7	1337	This work
		–	22.2	16.7	1340	[19]
$Co_5Er \leftrightarrow Co_{17}Er_2 + Co_7Er_2$	Eutectoid	16.7	10.5	22.2	1240	This work
		16.7	10.5	22.2	1240	[19]
$L + Co_3Er \leftrightarrow Co_7Er_2$	Peritectic	18.4	25	22.2	1352	This work
		–	25	22.2	1356	[19]
$L \leftrightarrow Co_3Er$	Congruent melting		25		1397	This work
			25		1395	[19]
$L + Co_3Er \leftrightarrow Co_2Er$	Peritectic	33.3	25	33.3	1338	This work
		–	25	33.3	1347	[19]
$L \leftrightarrow Co_2Er + Co_7Er_{12}$	Eutectic	58.5	33.3	63.2	801	This work
		56.25	33.3	64	795	[19]
$L \leftrightarrow Co_7Er_{12}$	Congruent melting		63.2		823	This work
			64		~825	[19]
$L \leftrightarrow Co_3Er + Co_7Er_{12}$	Eutectic	66	63.2	75	814	This work
		67.5	64	75	805	[19]
$L + (Er) \leftrightarrow CoEr_3$	Peritectic	74	100	75	911	This work
		–	100	75	912	[19]

According to the description of the $Co_{17}Er_2$ phase from Okamoto and Massalski [20], the calculated results are considered to be acceptable. All the invariant reactions in the Co–Er system compared with the available experimental data are listed in Table 4.

The calculated enthalpies of formation at 825 °C and 700 °C for the intermetallic compounds in the Co–Er system together with the experimental data are shown in Fig. 4. The reference states of pure elements of Co and Er are fcc and hcp phases, respectively.

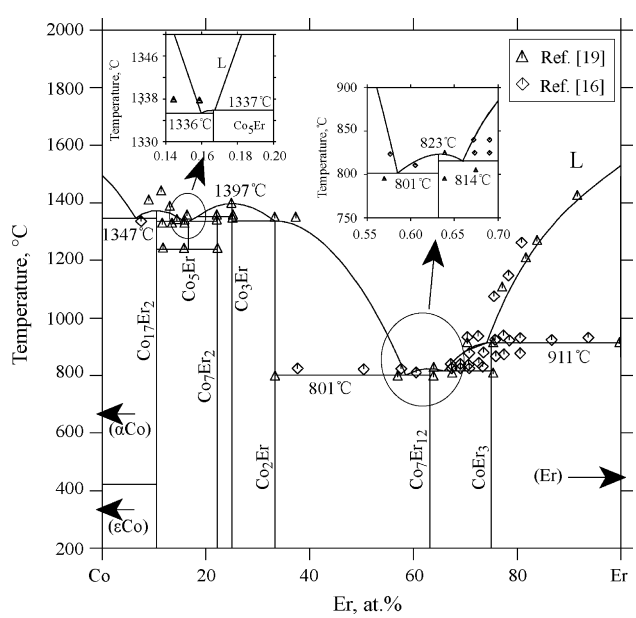

Fig. 3. The calculated phase diagram of the Co–Er system compared with the experimental data.

5.2. The V–Er system

The calculated phase diagram of the V–Er system is shown in Fig. 5 together with the experimental information [29]. According to the calculated results by the optimized parameters, the extrapolated miscibility gap of the liquid phase was predicted in the V–Er system, as shown in Fig. 5 with dotted lines. The calculated enthalpy of mixing of the liquid phase at 2527 °C is shown in Fig. 6, which shows the smaller values in the whole compositional range. The

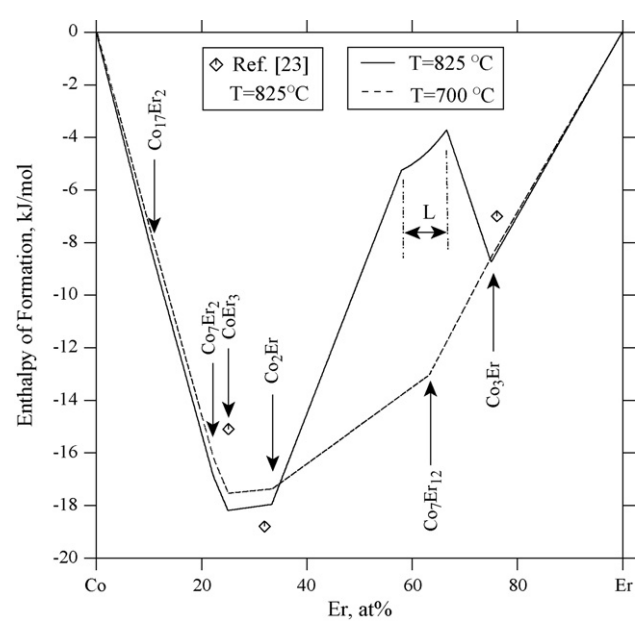

Fig. 4. The calculated enthalpies of formation at 825 °C and at 700 °C in the Co–Er system compared with the experimental data [23]. Reference state: fcc phase for Co and hcp phase for Er.

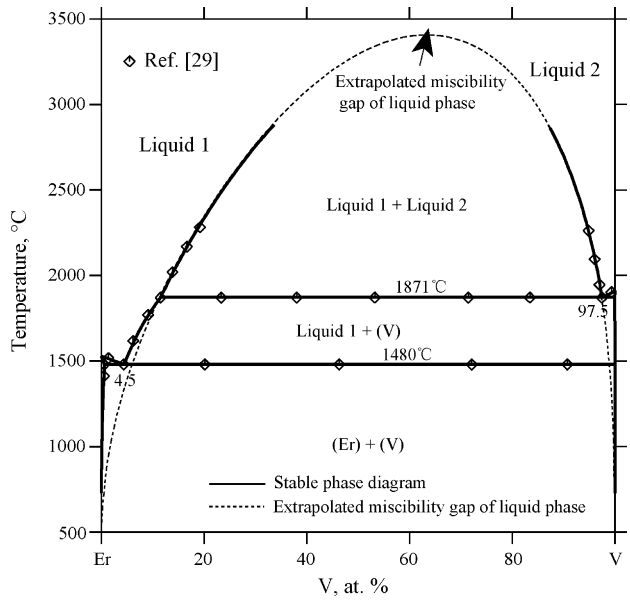

Fig. 5. The calculated phase diagram of the V–Er system with the extrapolated miscibility gap of the liquid phase.

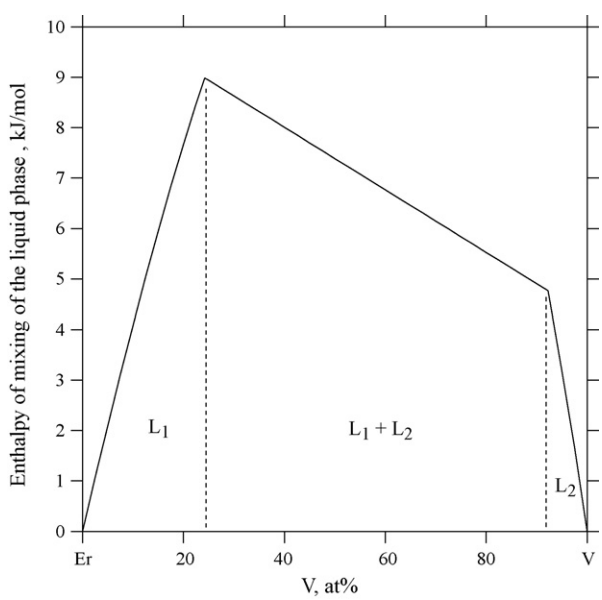

Fig. 6. The calculated enthalpy of mixing of the liquid phase at 2527 °C in the V–Er system. Reference state: liquid phase for V and Er.

Table 5
Calculated invariant reactions in the V–Er system compared with the data in Ref. [29].

Reaction	Reaction type	Composition at% V			T (°C)	Ref.
L1 ↔ L2 + (V)	Monotectic	11.4	97.5	99.9	1870	[29]
		11.5	97.5	99.9	1871	This work
L ↔ (Er) + (V)	Eutectic	4.3	0.4	99.99	1480	[29]
		4.5	0.6	100	1480	This work

calculated invariant reactions compared with the experimental data are listed in Table 5.

In the present assessment, all thermodynamic parameters for expressing the Gibbs free energies of reference states of pure elements are in accordance with SGTE pure element database [9].

6. Summary

The phase diagrams and thermodynamic properties in the Co–Er and V–Er binary systems were calculated by combining the thermodynamic models with the available experimental data in the literatures. A consistent set of the thermodynamic parameters has been obtained for each phase in the Co–Er and V–Er binary systems, and a reasonable agreement is obtained between the calculated results and experimental data.

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